



ALBUKHARY INTERNATIONAL UNIVERSITY



ASSIGNMENT: GROUP PROJECT

CCS2263- CYBERSECURITY SYSTEMS

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Table of Content

1. A network and device monitoring solution to enable the company to effectively detect threats and take actions.....	2
1.1 Benefits of the Proposed Solution.....	2
1.2 Solution Overview.....	3
2. Using computer forensic technique.....	3
2.1 Objectives of the Forensic Process.....	3
2.2 Forensic Process Overview.....	4
2.3 Case Example: SQL Injection on Patient Database.....	4
2.4 Importance of Forensic Readiness.....	5
3.A Secure Solution to Communicate with the Branches.....	5
4.Protection of the Company's Internal LAN.....	6
5. Identify and exploit vulnerabilities in the system to understand its weaknesses and improve security using penetration testing.....	6
5.1 Penetration Testing Phases.....	6
5.2 Tools Used for Penetration Testing.....	7
5.2.1 Kali Linux.....	7
5.2.2.Nmap.....	7
5.2.3 SQLMap.....	8
5.3 Vulnerability Management.....	8
6. VIDEO LINK OF THE PROJECT :	
https://drive.google.com/drive/u/2/folders/1ILKWGVUbSSLBY-XjK5LIF98Z3ndD8FHH.	8

1. A network and device monitoring solution to enable the company to effectively detect threats and take actions

With healthcare data being a prime target for cybercriminals, MyMediTech faces serious threats that can lead to data breaches, reputational harm, and regulatory consequences. Moreover, the geographically distributed infrastructure increases the complexity of security management. A comprehensive monitoring solution is essential to:

- Detect security threats in real-time
- Prevent unauthorized access and breaches
- Ensure availability, confidentiality, and integrity of healthcare data
- Maintain compliance with Malaysian PDPA and HIPAA-like standards

1.1 Benefits of the Proposed Solution

The proposed monitoring system offers both operational and security advantages, including:

- Real-time monitoring of network traffic with intelligent alert systems
- Device health tracking (CPU, memory usage, etc.)
- AI-powered anomaly detection and behavioral analysis
- A centralized dashboard for IT and security management
- Seamless integration with existing hospital and cloud systems

Recommended tools for implementation include open-source solutions like Nagios, Zabbix, or Wazuh, as well as commercial platforms such as SolarWinds, PRTG, or ManageEngine OpManager. The selection depends on budgetary considerations and desired customization capabilities.

1.2 Solution Overview

Our solution aims to significantly enhance system reliability and data security for MyMediTech. Key operational improvements include:

- Minimizing system downtime
- Ensuring 24/7 service availability
- Faster troubleshooting with centralized alerting
- Automating responses to known threats

From a security perspective, it:

- Strengthens compliance with PDPA and HIPAA
- Maintains audit trails for accountability and reporting
- Enables proactive risk mitigation to prevent ransomware and data leakage

2. Using computer forensic technique

2.1 Objectives of the Forensic Process

Our approach has three main objectives:

1. **Investigate Breached Systems:** We aim to determine how the attack happened—what systems were affected, which files were altered, and how the attacker infiltrated our environment.
2. **Recover Deleted or Hidden Evidence:** Attackers often cover their tracks by deleting or hiding files. We use advanced tools to recover these deleted files and reconstruct the sequence of activities.
3. **Ensure PDPA-Complaint Forensic Procedures:** As a healthcare technology company, it is essential that our forensic activities adhere to Malaysia's Personal Data Protection Act (PDPA) to ensure compliance with legal standards and ethical practices.

2.2 Forensic Process Overview

The following steps outline our forensic investigation workflow, which is applied each time a breach is suspected:

1. **Detect Breach:** The first step is identifying abnormal activities using our Security Information and Event Management (SIEM) system. This could involve unusual login patterns or abnormal data transfers.
2. **Create Forensic Disk Image:** Once a breach is suspected, we make a bit-for-bit copy of the affected machine's drive using tools like FTK Imager. This step ensures the preservation of original evidence.
3. **Load Image in Autopsy:** The disk image is then imported into Autopsy, a forensic tool that allows us to examine system artifacts, logs, and deleted files.
4. **Analyze Logs, Deleted Files, USB Devices, and Timestamps:** During this phase, we focus on identifying traces of attacker behavior, such as accessed files, deleted items, and evidence of external devices like USBs being used.

5. **Generate Evidence Report:** The findings from the analysis are compiled into a formal forensic report, which is crucial for both internal reviews and potential legal proceedings.

2.3 Case Example: SQL Injection on Patient Database

In a real-world scenario, MyMediTech responded to a suspected SQL injection attack on its patient database (`patients.db`). The breach was identified due to suspicious queries and system slowdowns. Here's how we applied our forensic process:

1. **Isolation and Cloning:** The first step involved isolating and cloning the affected server's drive to ensure evidence integrity.
2. **Forensic Analysis:** The forensic image was loaded into Autopsy. During the analysis, we discovered a deleted script, `sql_exploit.php`, and found traces of a USB device being inserted shortly after the breach.
3. **External Access Detection:** Our investigation revealed that the attacker accessed the database via an external IP address, which further confirmed the breach's scope.

A timeline of key events such as file deletion and USB insertion is shown in the Autopsy interface. This case highlights the effectiveness of our forensic process in uncovering critical evidence.

2.4 Importance of Forensic Readiness

The forensic process offers several benefits:

1. **Legal Defensibility and Chain of Custody:** Strict adherence to forensic procedures ensures that all digital evidence can be used in court or during compliance audits. Evidence is handled without modification, preserving its integrity.
2. **Reconstructing the Attack Timeline:** Our forensic process helps build a clear timeline of the attacker's actions, detailing when the breach started, how it progressed, and the subsequent actions taken by the attacker.
3. **Ensuring PDPA Compliance:** The process also ensures that sensitive personal data is managed responsibly, supporting internal audits and compliance with the PDPA.
4. **Prevention of Future Incidents:** Each investigation reveals vulnerabilities, guiding us in strengthening our security posture to prevent similar incidents from occurring.

3.A Secure Solution to Communicate with the Branches

To ensure secure and reliable communication between the headquarters and branch offices, a Site-to-Site VPN is proposed. This solution allows private, encrypted, and authenticated data exchange across all company locations over the internet. The VPN utilizes IPsec

protocols, implementing strong encryption algorithms such as AES-256 and SHA-2 to maintain confidentiality and integrity.

Deployment can be done using firewalls like pfSense, Cisco ASA, or FortiGate, with authentication through digital certificates or pre-shared keys. For flexible deployment or performance considerations, OpenVPN or WireGuard may also be considered.

To enhance overall communication security, it is also recommended to use:

- End-to-end encrypted messaging platforms (e.g., Signal, Matrix),
- TLS-secure email systems (e.g., ProtonMail, Postfix),
- Multi-Factor Authentication (MFA) for all VPN connections.

Supporting security practices include the auto-renewal of VPN credentials, centralized log monitoring using platforms like Wazuh or the ELK Stack, and the adoption of Zero Trust principles, ensuring that only verified users and devices are allowed access.

This comprehensive strategy guarantees that even when using public networks, all inter-branch communications remain encrypted, authenticated, and fully protected.

4. Protection of the Company's Internal LAN

Internal Local Area Network (LAN) protection is one of the essential elements to protect the organization against internal attacks and block lateral transactions of an attacker. This security starts at the network level by segregating the sections with VLANs into departments like HR, Finance, and IT departments so as not to be accessed unnecessarily. Access Control Lists (ACLs) are used to enforce an inter-VLAN interconnection in a secure manner.

Intrusion Detection and Prevention Systems (IDPS) like Snort or Suricata are installed in order to monitor living and block suspicious traffic. Intrusion detection systems are able to identify irregularities such as port scans, brute-force logs, and malware symbols at the present moment.

At the end-point level, EDR tools like CrowdStrike or Microsoft Defender ATP offer behavioral analysis, real-time notifications as well as automated response to take infected devices off the network.

Additional security measures include:

- Multi-Factor Authentication (MFA) across all user accounts,
- Regular patching through tools like WSUS or Ansible,
- Antivirus solutions with threat hunting capabilities,
- Centralized log management using Wazuh or other SIEM systems.

This layered security architecture strengthens internal defenses, ensuring that even if an attacker bypasses perimeter protections, further damage within the network is minimized or entirely prevented.

5. Identify and exploit vulnerabilities in the system to understand its weaknesses and improve security using penetration testing.

5.1 Penetration Testing Phases

Our penetration testing process at MyMediTech is divided into six key phases:

1. **Planning:** This is where we define which systems and applications to test, such as the hospital management portal. It's essential to establish clear testing boundaries and objectives.
2. **Information Gathering:** During this phase, we collect relevant details about the target systems. This involves identifying the architecture, technologies, and potential entry points of the system.
3. **Vulnerability Identification:** Here, we scan for vulnerabilities using industry-standard tools like Nmap and Burp Suite. This step helps us identify weak spots that attackers may exploit.
4. **Attack Execution:** We simulate real-world attacks to exploit vulnerabilities and understand how they could be leveraged by malicious actors. This phase is critical for understanding the potential impact of these weaknesses.
5. **Impact Assessment:** After simulating attacks, we assess their potential impact on the organization, especially with regard to the safety and confidentiality of patient data.
6. **Reporting and Remediation:** The final phase involves documenting all findings and suggesting fixes to mitigate identified vulnerabilities. These reports also help ensure compliance with relevant security standards.

5.2 Tools Used for Penetration Testing

To enhance the penetration testing process, MyMediTech uses several specialized tools, each designed for specific testing tasks.

5.2.1 Kali Linux

Kali Linux is a versatile operating system specifically built for penetration testing and ethical hacking. It comes pre-loaded with numerous tools for tasks such as information gathering, vulnerability scanning, and web application testing. At MyMediTech, Kali Linux is used to simulate real cyberattacks, helping us identify weaknesses before they can be exploited by actual attackers.

5.2.2.Nmap

Nmap is a powerful network scanning tool used to identify active hosts, open ports, and possible vulnerabilities in a network. We use Nmap (through its graphical version, Zenmap) to visually map out devices and identify exposed systems on our network. The tool's command-line version is also used for specific vulnerability detection, such as identifying backdoor vulnerabilities in FTP services.

5.2.3 SQLMap

SQLMap is an essential tool for detecting and exploiting SQL injection vulnerabilities in web applications. Considering that MyMediTech handles sensitive healthcare records, SQLMap helps us ensure that our backend databases are secure against attacks that might otherwise allow malicious actors to access confidential patient data.

5.3 Vulnerability Management

After performing penetration testing, MyMediTech follows a structured vulnerability management process to further improve system security. The process includes:

1. Pre-assessment: Identifying critical assets and understanding the scope of the testing.
2. Vulnerability Scanning: Using the results from penetration testing to conduct in-depth vulnerability scans.
3. Risk Assessment and Remediation: Prioritizing vulnerabilities based on their impact and severity, and taking action to patch or mitigate risks.
4. Verification: Ensuring that the identified vulnerabilities have been addressed and the patches are effective.
5. Ongoing Monitoring: Continuously assessing the system and running periodic scans to identify any new vulnerabilities.

This lifecycle helps us ensure that we not only address current vulnerabilities but also improve internal processes to make the system more secure over time.

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